Project Report

On

SMART GROCERY

Created By

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Bscit-6

Under The Guidance of

PROF.BRIJESH MARU

Submitted To

GEETANAJALI COLLEGE OF COMPUTER SCIENCE & COMMERCE

Affiliated To

Saurashtra University

For Academic Year

2025-26

Bytes Bizz Technology
Gateway to Your Business

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that **Chauhan Ravi ID:** BT0250085 **Geetanjali College, Rajkot** has successfully completed the **PHP Development Internship** with **Bytesbizz Technology**. The internship was conducted **offline** from **22nd December 2025 to 4th February 2026**. During this period, Ravi was assigned various tasks and projects aimed at enhancing his practical knowledge and skills in PHP development and web technologies.

Internship Description & Learning Outcomes:

- Gained hands-on experience in **PHP programming and server-side web development**.
- Worked on **real-world web application projects**, implementing dynamic and interactive features.
- Learned **database integration, backend logic handling, and basic performance optimization techniques**.
- Enhanced **problem-solving skills, analytical thinking, and professional work ethics**, preparing for industry-level challenges.

Ravi demonstrated dedication, sincerity, and professionalism throughout the internship period. We hereby acknowledge his successful completion of the internship and wish him every success in his future endeavors.

For Bytesbizz Technology,

Mr. Yash Buhecha,
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GEETANJALI COLLEGE

Of Computer Science & Commerce (B.B.A.)

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Ref. No.

Date 06/11/2025

Roll No.

Seat No : 23SI00 3UG00139

Certificate

This is to certify that Mr./Miss. Chauhan Ravi G. _____ student of B.Sc. I.T. SEM-V has carried out this project work as per Saurashtra University curriculum, under our guidance and supervision and it is his/her own contribution.

PROJECT GUIDE	I/C PRINCIPAL
Geetanjali College Of Computer Science & Commerce (BBA)	Geetanjali College Of Computer Science & Commerce (BBA)

REDMI 9 PRIME
AI QUAD CAMERA

Acknowledgement

- I would like to express my sincere gratitude to **Bytes Bizz Technology** for providing me with the opportunity to develop the **Smart Grocery System** using **PHP**. Working on this project has been a valuable learning experience that allowed me to apply my knowledge of web technologies such as **PHP, MySQL, HTML, CSS, and JavaScript** in a practical environment.
- I am highly thankful to the mentors and team members at Bytes Bizz Technology for their continuous guidance, support, and encouragement throughout the development of this project. Their technical advice and constructive suggestions helped me understand the concepts of web application development, database management, and system design.

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INTRODUCTION

1. OBJECTIVE

- The main objective of this project is to develop an **online Smart Grocery Store System** that simplifies grocery store management and online shopping for customers.
- The system allows customers to **register, log in, browse grocery products, add items to the cart, and place orders online** easily.
- It enables administrators to **manage product categories, products, orders, customers, and generate sales reports efficiently**.
- The system aims to provide a **fast, secure, and user-friendly platform** for both customers and administrators.

2. Scope of the Project

This project aims to **automate and digitalize grocery store operations**, such as:

- Customer **registration and authentication** for secure access.
- **Online grocery browsing and shopping cart management**.
- **Order placement and order tracking** for customers.
- **Administrative control** over product categories, product listings, and order management.
- **Inventory and sales management** through the admin panel.
- **Automated generation of sales and revenue reports**.
- **Secure login system** for both administrators and customers.

POJECT PROFILE

Project Title :- Smart Grocery

Organization :- Developed at **Bytes Bizz Technology**

Development Software : **Visual Studio**

Technologies Used

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL
- **Server:** Apache (XAMPP / WAMP)
- **Browser:** Google Chrome / Microsoft Edge / Firefox

Academic Year : - **2025-26**

Developed By : - **Ravi Chauhan**

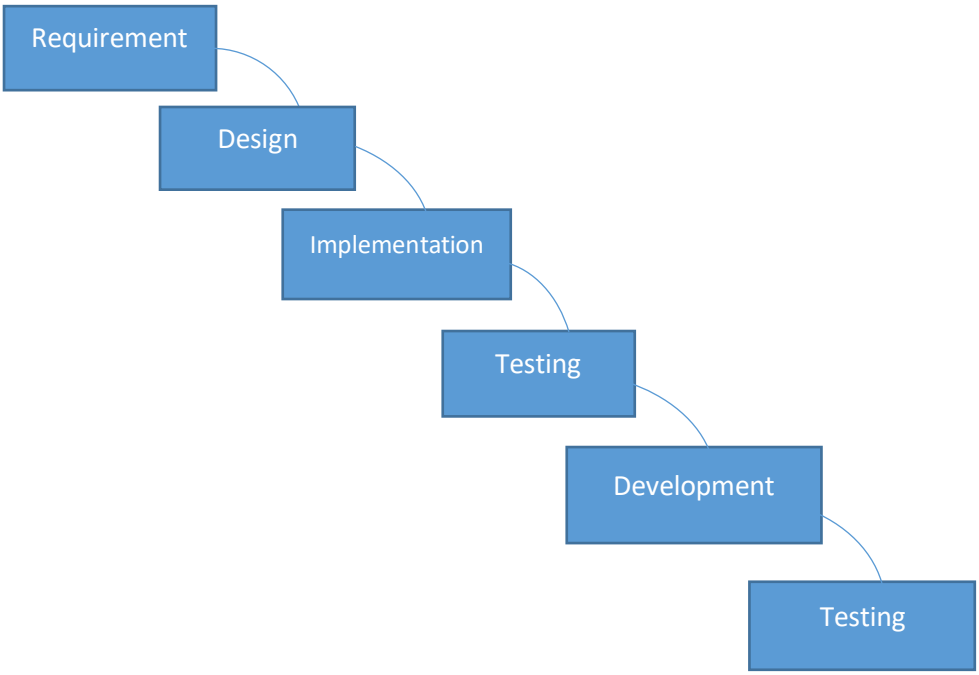
Submitted To : - **Geetanjali College**

PROCESS MODEL

- For The development of the Online Food Ordering System, the Waterfall Model has been used.

The Waterfall Model:

- The **Waterfall Model** is one of the most widely used **Software Development Life Cycle (SDLC)** models that follows a **sequential design process**. In this model, each phase of development must be completed before moving to the next phase.
- For the **Smart Grocery Store System**, the Waterfall Model helps in organizing the development process in a systematic way. The project begins with **requirement analysis**, where all the system requirements such as customer registration, product browsing, shopping cart, order placement, and admin management are clearly identified.



Advantages of Using Waterfall Model

- Easy to Understand and Manage.
- Each phase has clear deliverables and deadlines.
- Works well when requirements are fixed and clearly defined.
- Simple and structured approach.

Software Resources

Frontend Design Tool : Visual Studio Code

Database : Microsoft SQL server

Code-Behind Language : PHP

Other Technologies Used: HTML, CSS, JavaScript

Customer Module Functionality

The **Customer Module** allows users to interact with the Smart Grocery Store system and purchase products online.

- **User Registration** – Customers can create a new account by providing personal details.
- **Login / Logout** – Registered users can securely log in and log out of the system.
- **View Products / Grocery Items** – Customers can browse different grocery categories and products.
- **Add to Cart** – Customers can add selected grocery items to their shopping cart.
- **Update / Remove Cart Items** – Users can modify the quantity of items or remove items from the cart.
- **Checkout** – Customers can confirm their order and place it online.
- **View Order History** – Customers can view details of their previous orders.

Admin Module Functionality

The **Admin Module** allows the administrator to manage the entire Smart Grocery Store system.

- **Login / Logout** – Secure login access for the administrator.
- **Dashboard Overview** – Displays summary information such as total products, categories, users, and orders.
- **Add Product** – Admin can add new grocery products to the system.
- **Manage Products (Update / Delete)** – Admin can update product information or remove products.
- **Manage Categories** – Admin can create and manage product categories.
- **Manage Orders** – Admin can view, update, and process customer orders.
- **Manage Users / Customers** – Admin can manage registered customers.
- **Generate Sales / Revenue Reports** – Admin can view reports related to store sales and revenue.

Data Dictionary

1. User Login Table

Columns	Datatype	Constraints
User_id	int	PrimeryKey(Ai)
name	Varchar(100)	Not null
phone	Varchar(100)	Not null
email	Varchar(100)	Not null
password	Varchar(100)	Not null
role	Varchar(100)	Not null
address	text	null
city	Varchar(50)	null
pincode	Varchar(10)	null

2.Admin Login Table

Columns	Datatype	Constraints
id	int	PrimeryKey(Ai)
name	Varchar(100)	Not null
email	Varchar(100)	Not null
password	Varchar(255)	Not null

3.Category Table

Columns	Datatype	Constraints
Category_id	int	PrimeryKey(Ai)
category_name	Varchar(100)	Not null
slug	Varchar(120)	Not null
image	Varchar(255)	Not null
status	tinyint	null
Created_at	timestamp	Not null

4.Contactus Table

Columns	Datatype	Constraints
id	int	PrimeryKey(Ai)
name	Varchar(100)	Not null
email	Varchar(100)	Not null
subject	Varchar(150)	Not null
message	text	Not null
Created_at	timestamp	Not null

5.Product Table

Columns	Datatype	Constraints
Product_id	int	Primerykey(Ai)
Category_id	int	Not null
Product_name	Varchar(100)	Not null
Short_description	Varchar(255)	Not null
price	Decimal(10,2)	Not null
mrp	Decimal(10,2)	Not null
Stock_qty	int	Not null
Image	Varchar(255)	Not null
description	text	Not null
status	tinyint	Not null

6.Delivery Boys

Columns	Datatype	Constraints
delivery_id	int	Primerykey(Ai)
name	Varchar(100)	Not null
phone	Varchar(15)	Not null
email	Varchar(100)	Not null
password	Varchar(255)	Not null
status	tinyint	Not null
Created_at	timestamp	Not null

7. Orders

Columns	Datatype	Constraints
Order_id	int	Primerykey(Ai)
Customer_id	int	Not null
name	Varchar(50)	Not null
phone	Varchar(50)	Not null
address	text	Not null
total	Decimal(65,2)	Not null
Collected_amount	Decimal(10,2)	Not null
Collected_at	Datetime(6)	null
Payment_method	Varchar(50)	null
Order_date	timestamp	null
status	enum	Not null
Address_id	int	null
Delivery_id	int	Not null
Delivery_status	enum	Not null
Delivery_otp	Varchar(6)	null
Delivered_at	datetime	null
Delivery_charge	Decimal(10,2)	null
savings	Decimal(10,2)	Noy null

8. Order Items

Columns	Datatype	Constraints
id	int	Primerykey(Ai)
Order_id	int	Not null
Product_id	int	Not null
quantity	int	Not null
image	Varchar(255)	Not null
price	Decimal(10,2)	Not null

9. Order Updates

Columns	Datatype	Constraints
id	int	Primerykey(Ai)
Order_id	int	Not null
Delivery_status	Varchar(100)	Not null
Payment_method	Varchar(50)	Not null
Collected_amount	Decimal(10,2)	Not null
Created_at	datetime	Not null

10. Product Images

Columns	Datatype	Constraints
Image_id	int	Primerykey(Ai)
Product_id	int	Not null
image	Varchar(255)	Not null
Created_at	timestamp	Not null

11. User Address

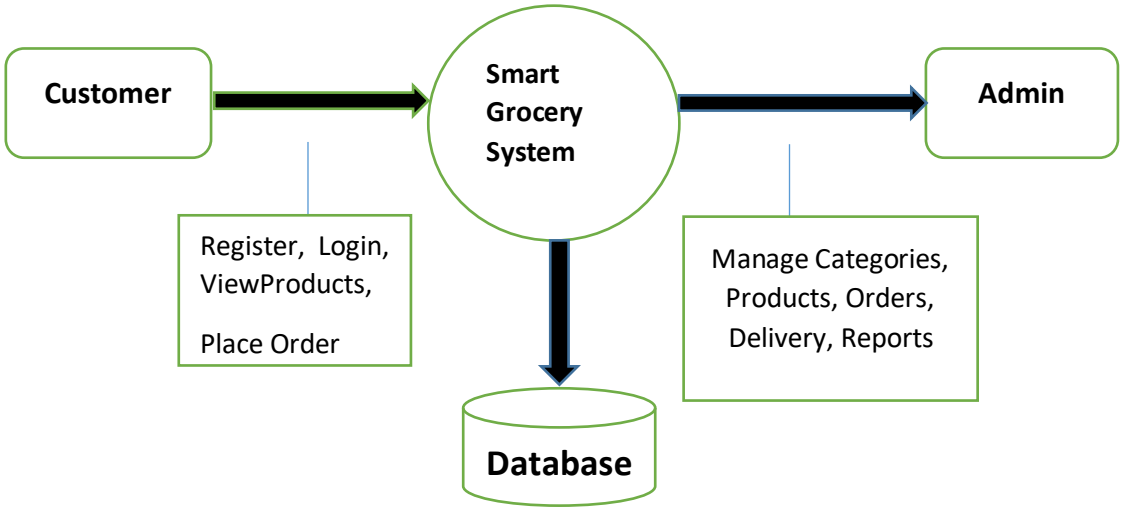
Columns	Datatype	Constraints
address_id	int	Primerykey(Ai)
user_id	int	Not null
name	Varchar(100)	Not null
phone	Varchar(15)	Not null
address	text	Not null
city	Varchar(50)	Not null
pincode	Varchar(10)	Not null
Is_default	tinyint	null
Created_at	timestamp	null

DFD(Data Flow Diagram)

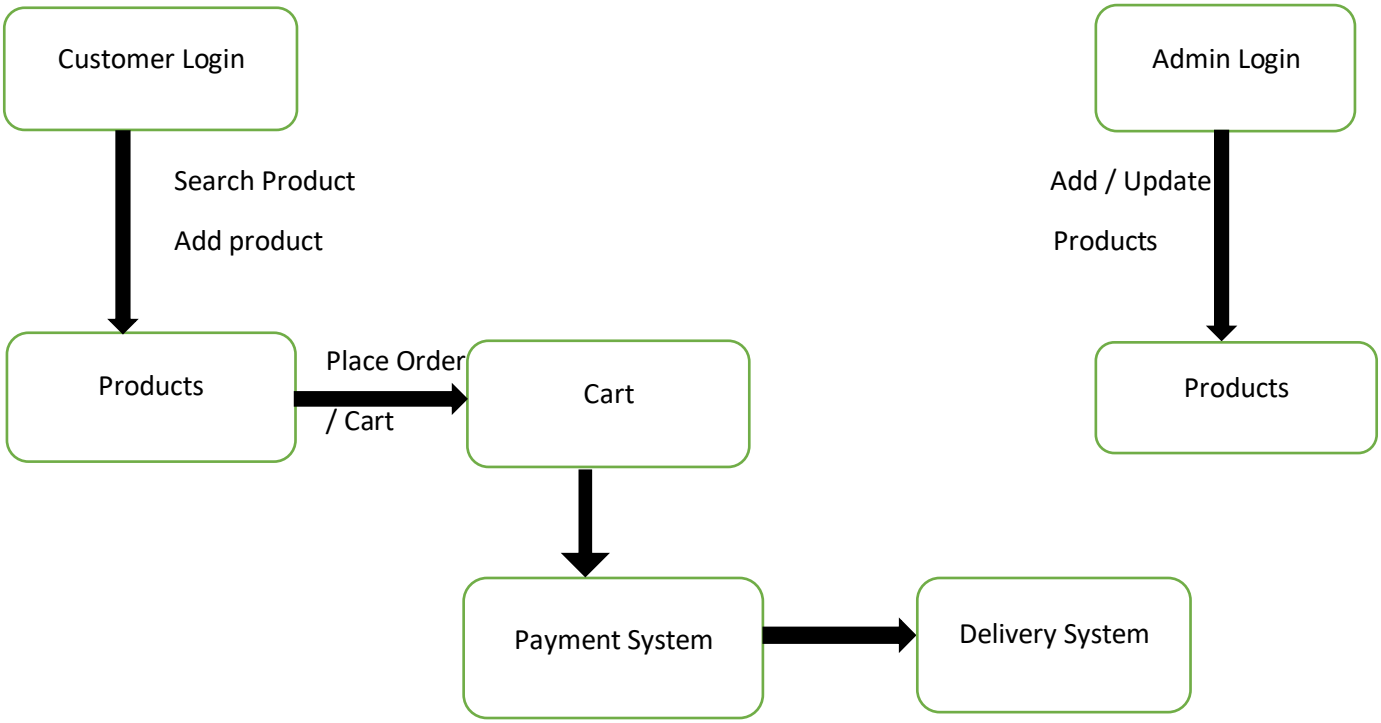
➢Data Flow Diagram Symbol

Symbol	Description
	Data Flow : data flow are pipelines through the Packets of information
	Entity : entity are the object of the system
	Process : A process or task performed by the system
	Data store : A palce where data to be stored

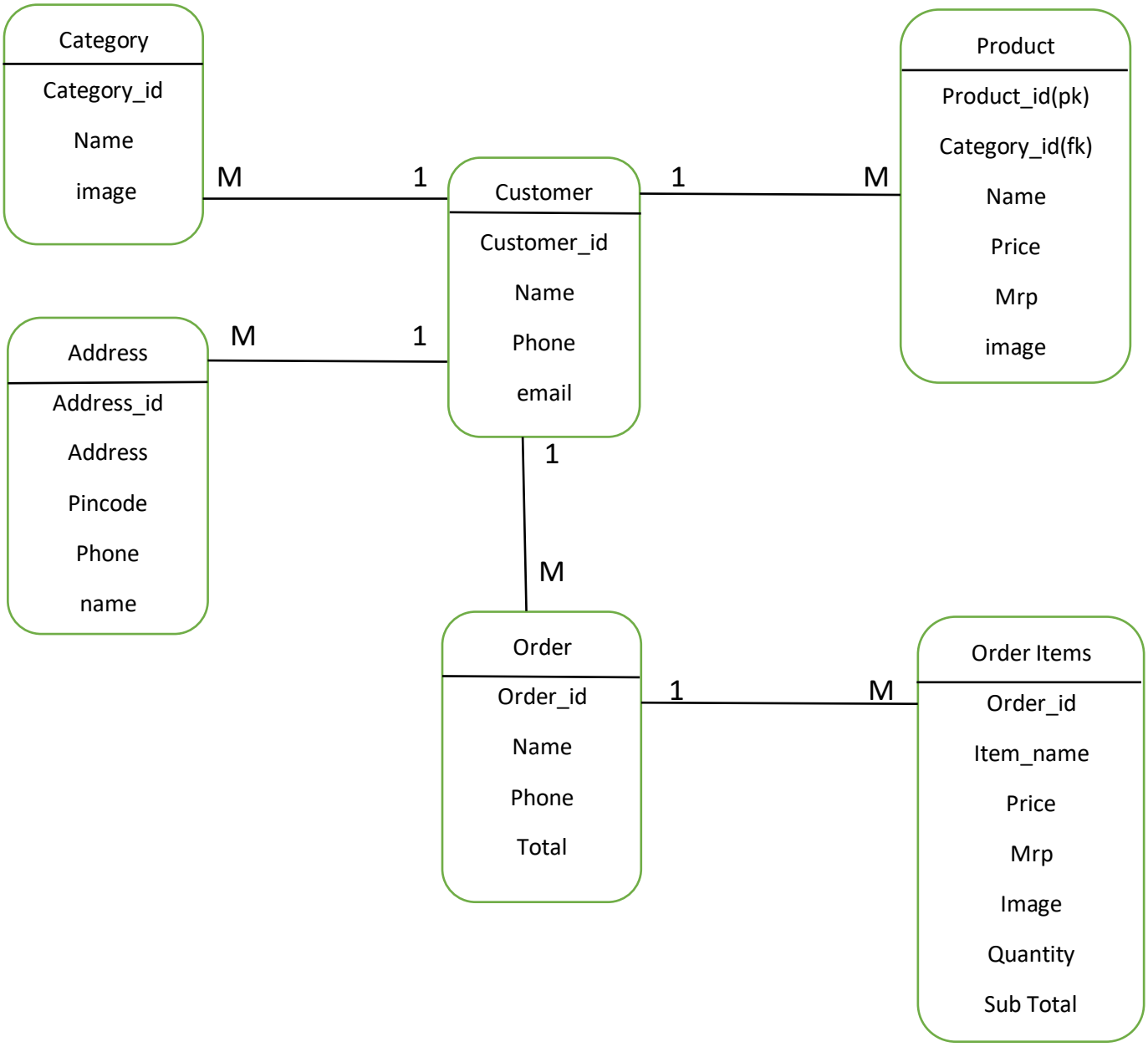
➢0 Level DFD



1 Level DFD


Customer Login
Search Product
Add product
Products
Place Order
/ Cart
Cart
Admin Login
Add / Update
Products
Products
Payment System
Delivery System

ERD [Entity Relationship Diagram]



Test Cases

User Registration

Serial No.	Test Case	Precondition	Result	Status
1.	Verify a new customer can register successfully	Registration page is accessible	User registered successfully and redirected to login page	Pass/Fail
2	Verify validation for exiting email or phone	Exiting user email or phone	System shows email or phone already exists	Pass/Fail

Login

Serial No.	Condition to be Check	Test Data	Expected Output	Remarks
1.	If Fields are empty	Value of form fields.	Alert the user to enter the fields and then proceed.	Pass/Fail
2	If incorrect username or password	Value of Textbox of Username & password	Alert user that “Invalid Username or password” Please Try Again	Pass/Fail

Cart

Serial No.	Test Case	Precondition	Result	Status
1.	Verify add to cart functionality	Select item to add to cart	Item added to cart successfully	Pass/Fail
2.	Update cart	Verify that user can update item quantity	Cart updated successfully	Pass/Fail
3.	Remove cart Item	Verifying Removing Item from cart	item removed from cart	Pass/Fail

Checkout

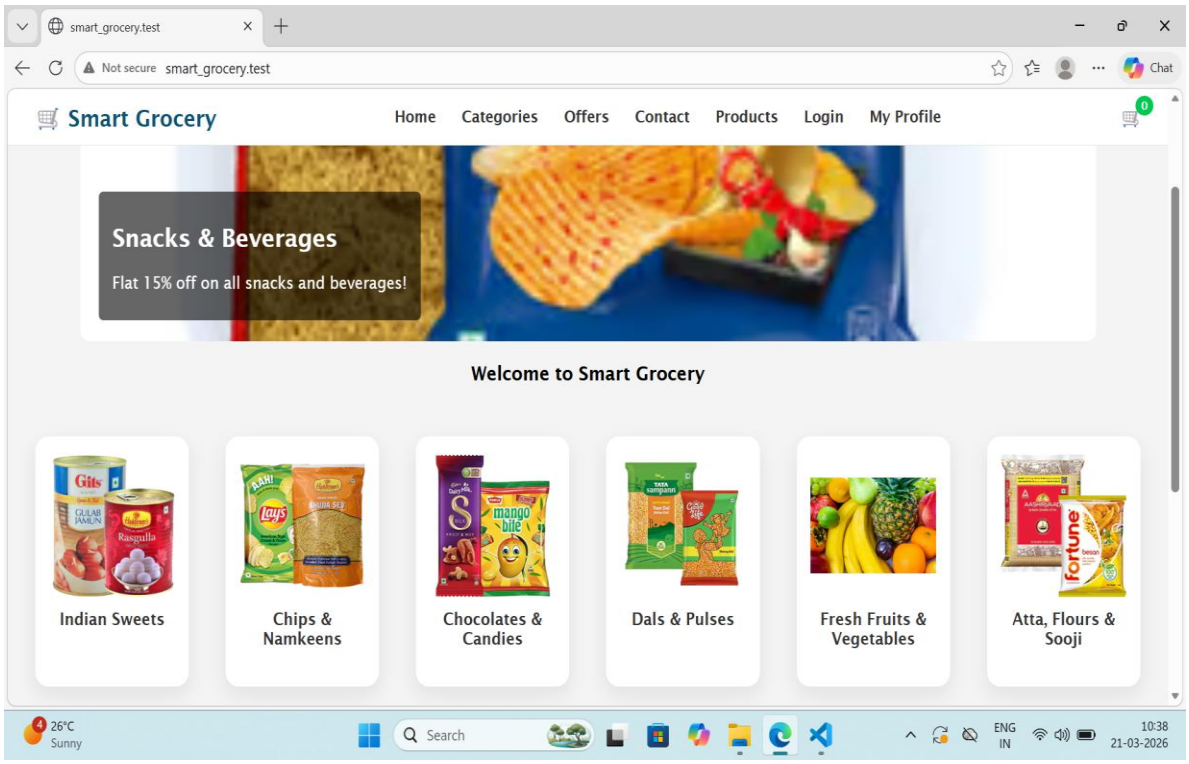
Serial No.	Test Case	Precondition	Result	Status
1.	At Least one item in cart	Add item to cart	Order Placed successfully	Pass/Fail
2	Select valid mobile number or address	Please Select valid phone or address	Order Placed successfully	Pass/Fail
3.	Select Payment method	Please select valid payment method to checkout cart items	Order Placed successfully	Pass/Fail
4.	Check if user Logged in	User Must be login for checkout	Order Placed successfully	Pass/Fail

Admin

Serial No.	Test Case	Precondition	Result	Status
1.	Add Category, Product	1.Admin Logged in. 2.all fields must be filled by admin .	Category/Product added successfully	Pass/Fail
2.	Update Category, Product	1.Admin Logged in. 2.all fields must be filled by admin . 3. click on Update	Category/Product updated successfully	Pass/Fail
3.	Delete Category, Product	1.Admin Logged in. 2. click on Delete	Category/Product deleted successfully	Pass/Fail
4.	Manage Orders	1.Admin Logged in. 2.click action dropdown 3. change Order status	Assigned Orders to Delivery Boys	Pass/Fail
5.	Revenue Report	1.Admin Logged in. 2.click on time period dropdown 3. click on generate.	Display Revenue Report of selected time Period	Pass/Fail
6	Logout	Admin must be loggedin	Redirected to Login/Home page session destroyed	Pass/Fail

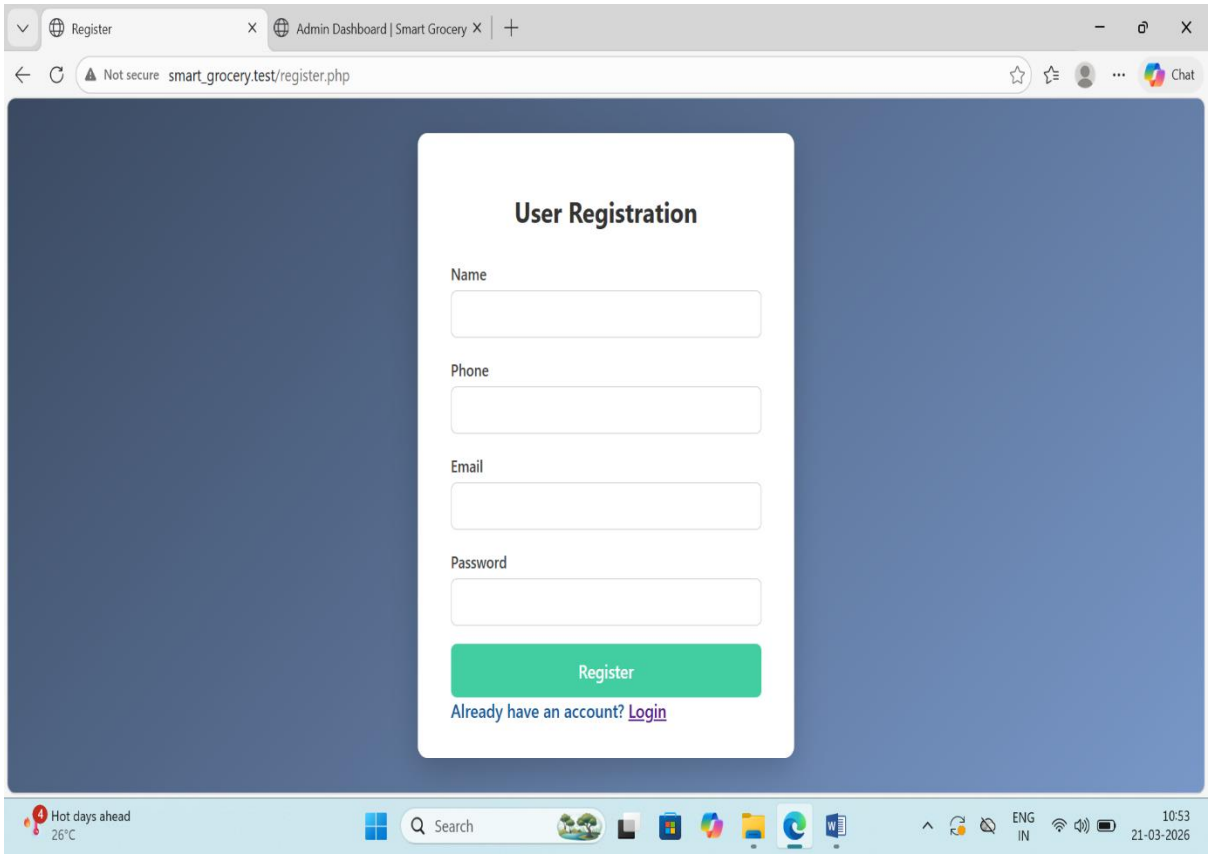
Screenshots

Home Page



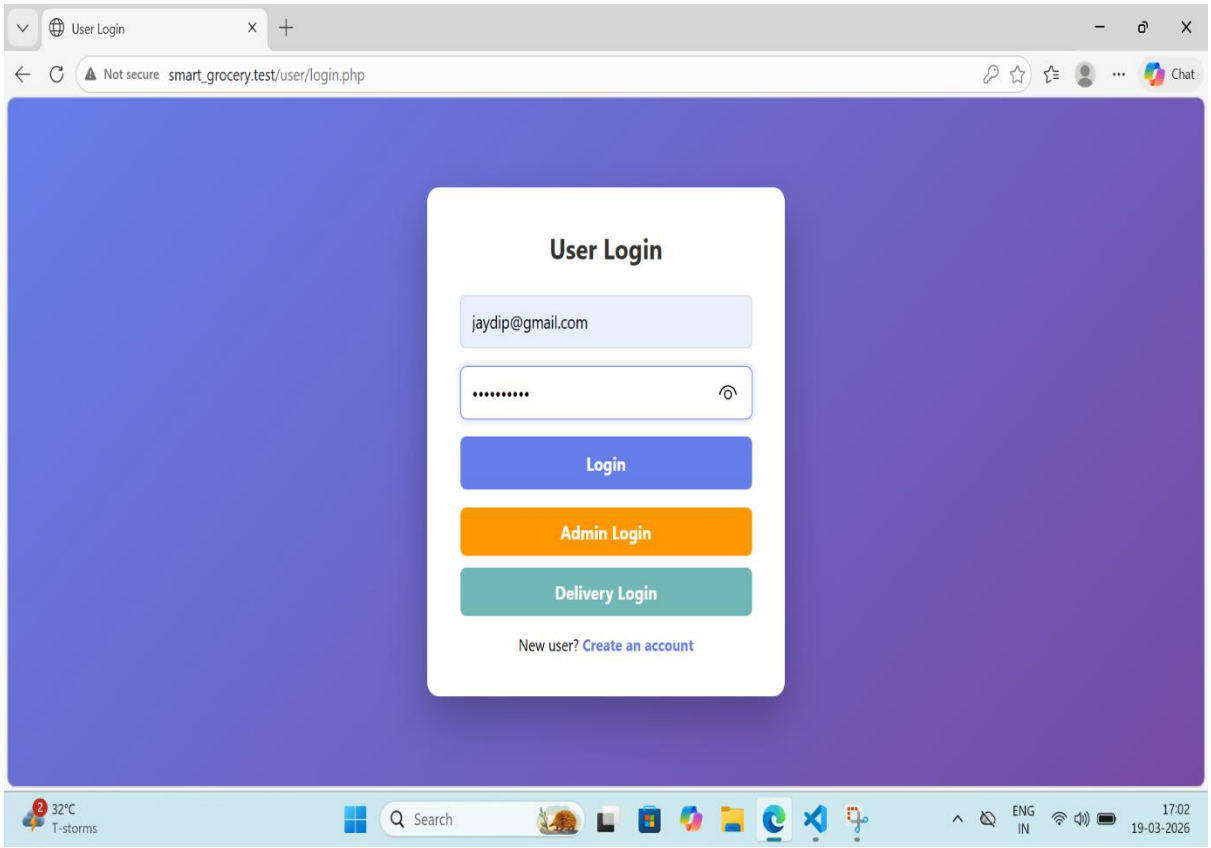
- Display Home page when user open The site.
- Form Home Page users can saw Multiple categories, Click on particular category for purchase something.

Register Page



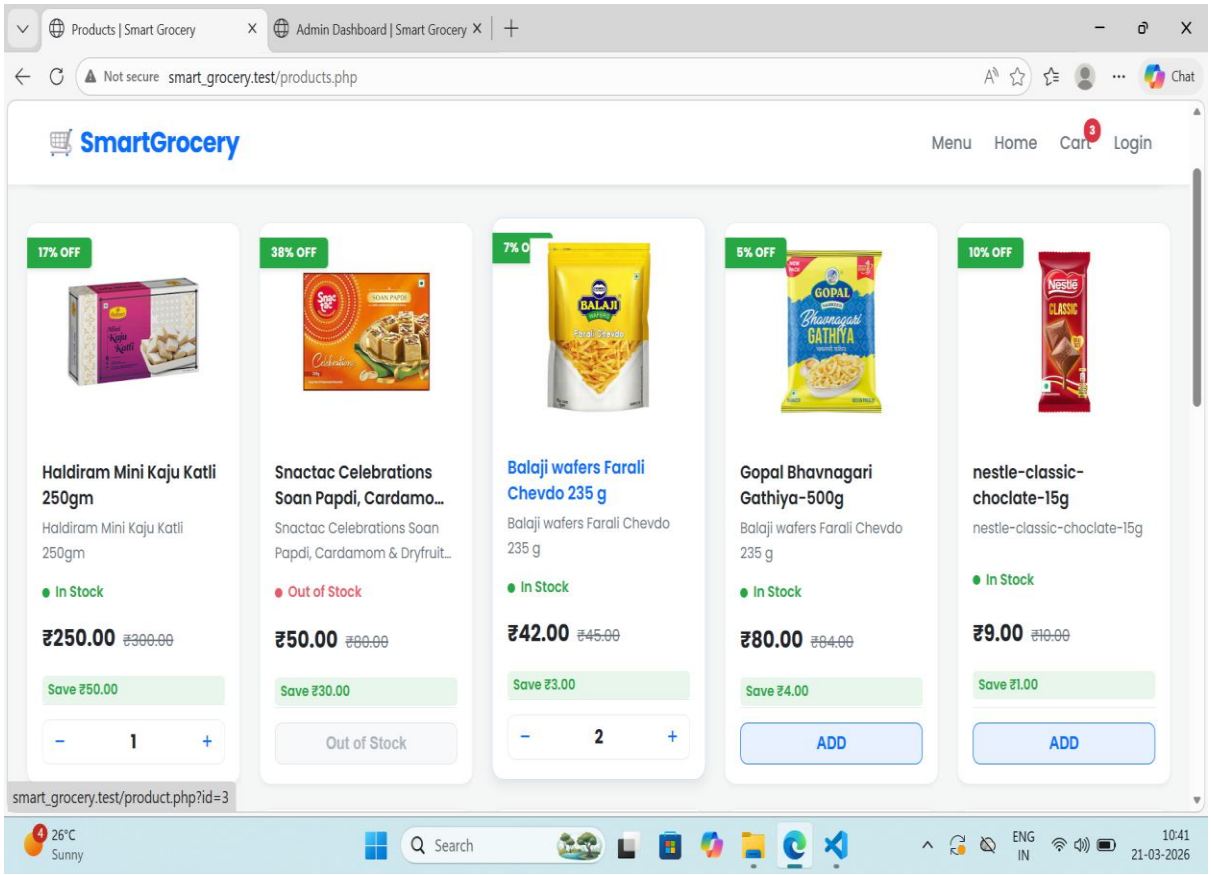
- Enter name, valid email, phone, Password, address to register.
- Allow user to create account .

Login Page



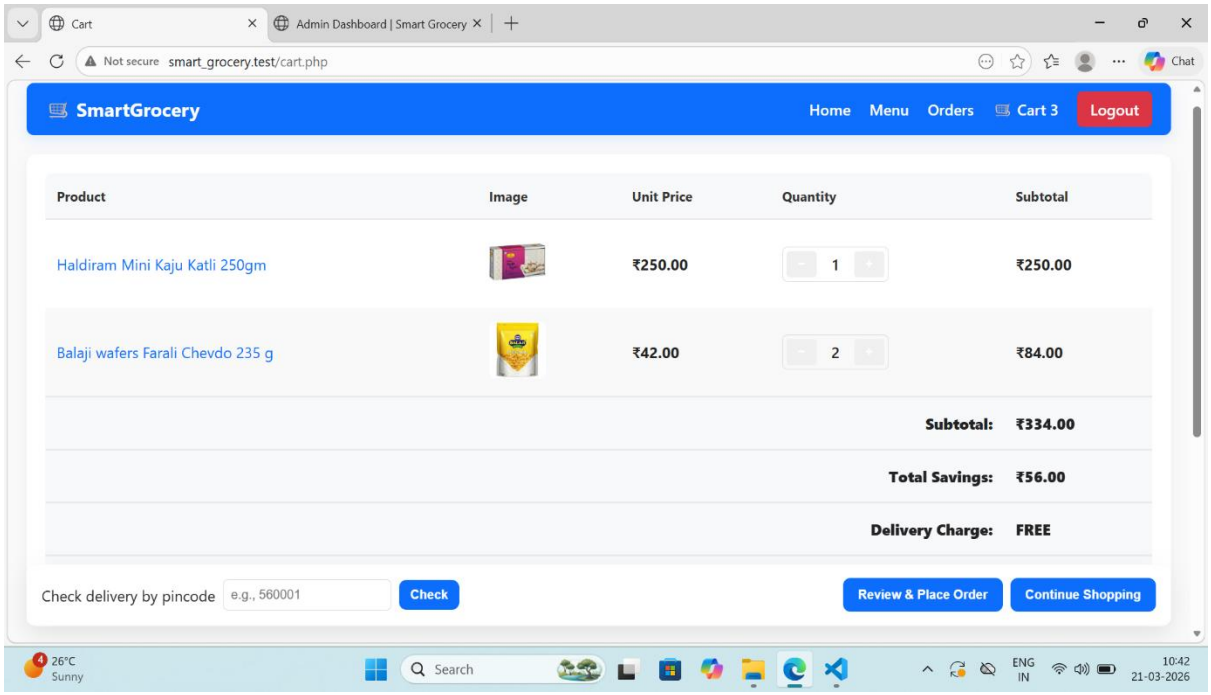
- For User Authentication.
- Login Only Registered Users.
- After Login customer redirect to home page and can give order and Admin can perform administrative tasks.

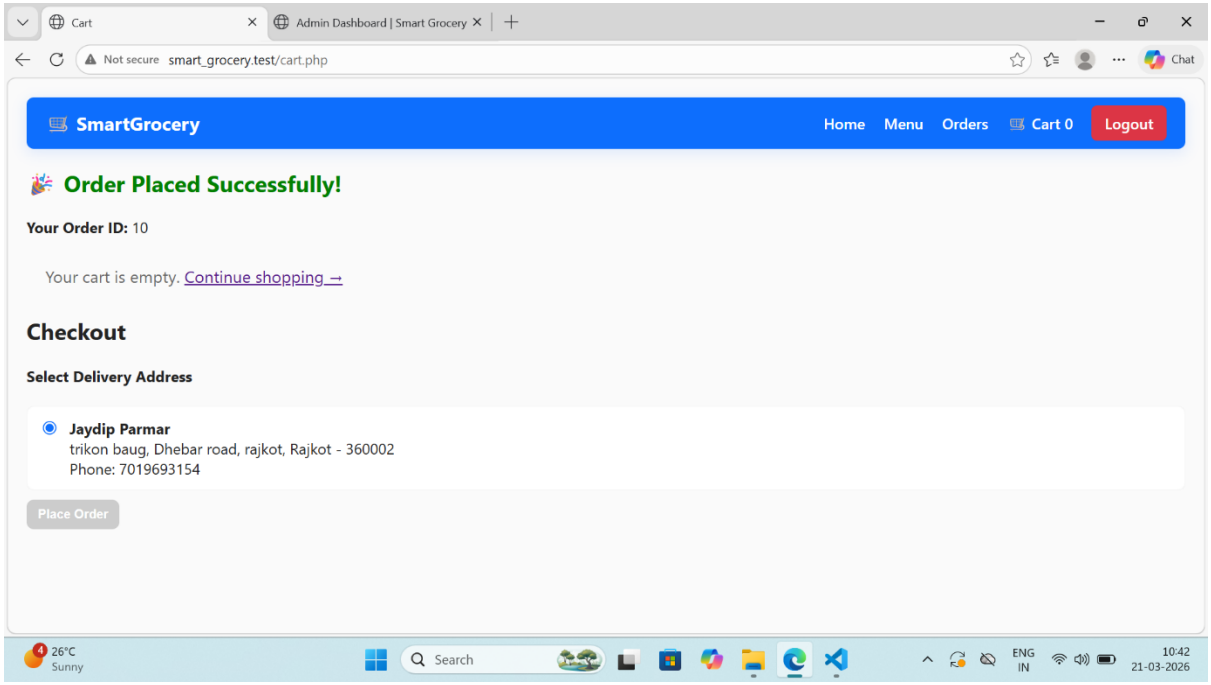
Products Page



- Customer can Search Particular Product by name or by category name.
- Customer can add minimum 1 item and maximum 6 or available stock.
- When item is out of stock add to cart button is disabled.

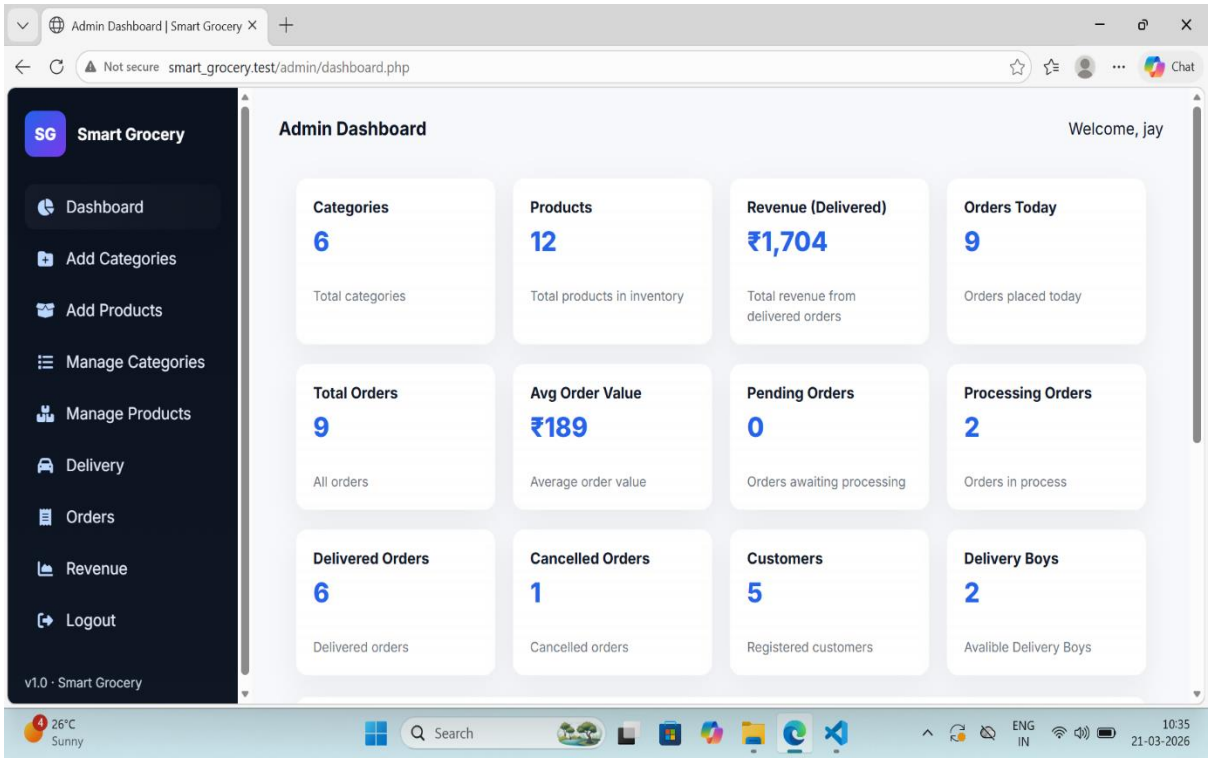
Cart Page





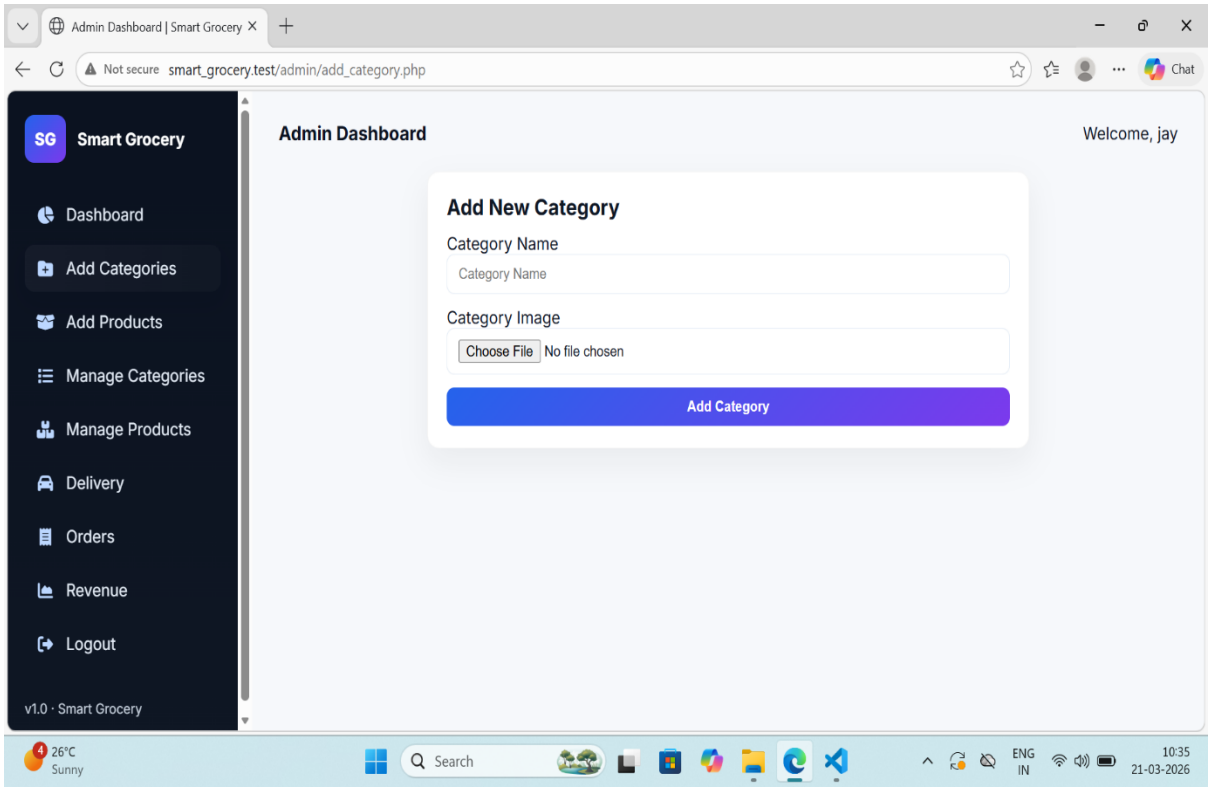
- Customer must be login for checkout cart items.
- Enter all fields for checkout.
- Please select payment method for checkout.

Admin Dashboard



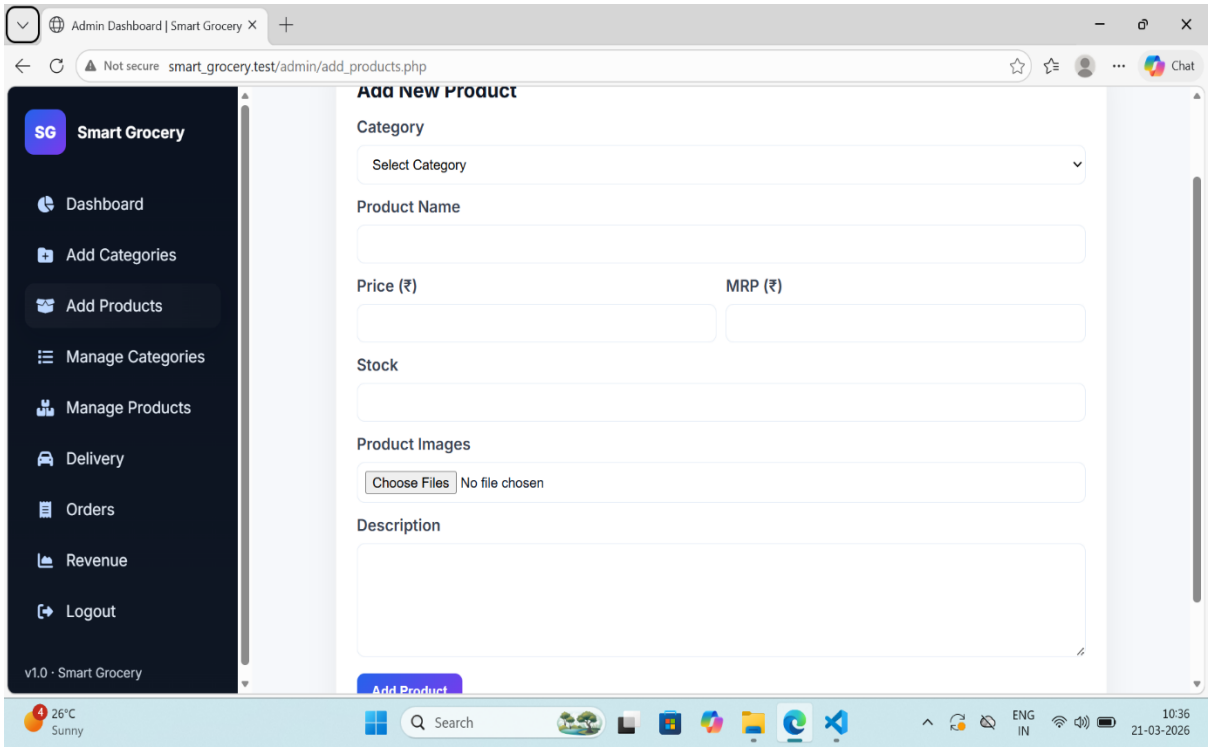
- This admin dashboard saw all administrative activities of the Smart Grocery System.
- Admin can perform all administrative from this Dashboard Like Add Categories, Add Products, Manage Categories, Manage Products, Assign Order to Delivery Boys, Generate Bill, Generate Revenue Report.

Add Category



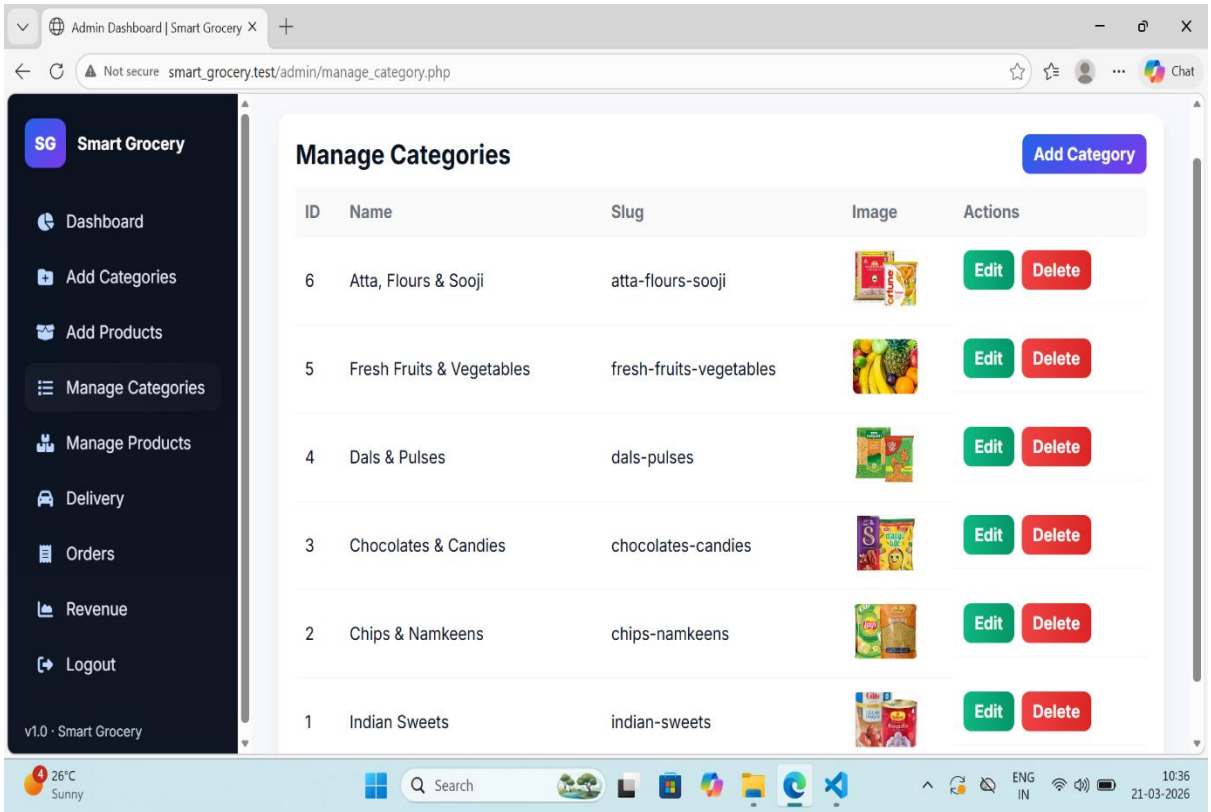
- Add category name and insert category image for add category.
- All Fields Must be filled out for add category.
- Choose valid image format of Category.

Add Product



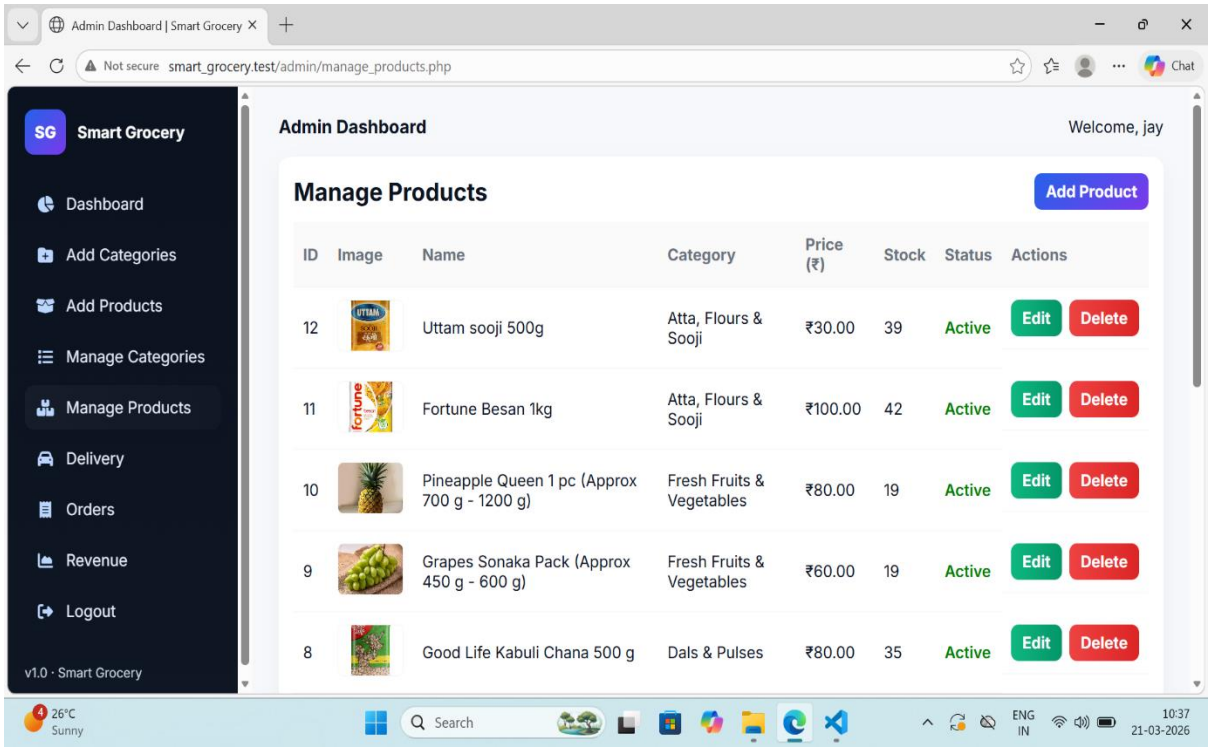
- Add Product name, price, mrp, description, stock and insert product image for add category.
- All Fields Must be filled out for add Product.
- Choose valid image format of Product.

Manage Categories



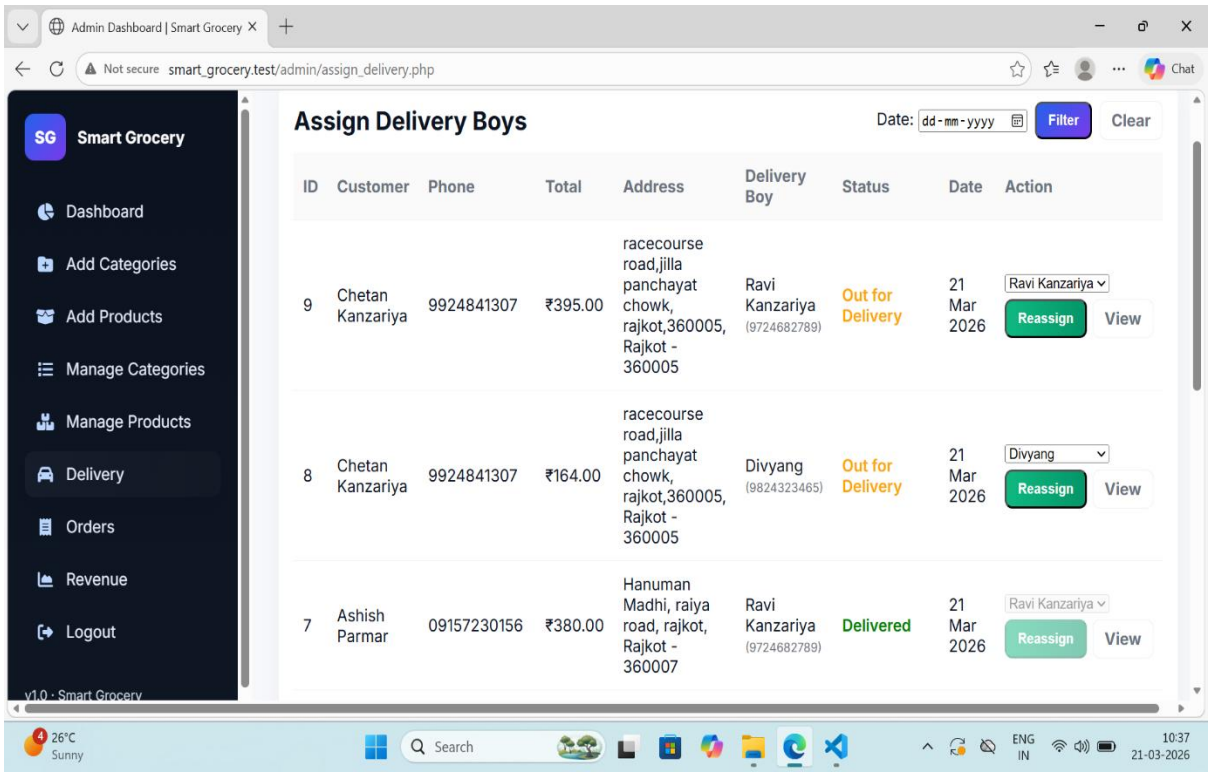
- Admin can Update and delete Category items from manage Categories page.

Manage Products



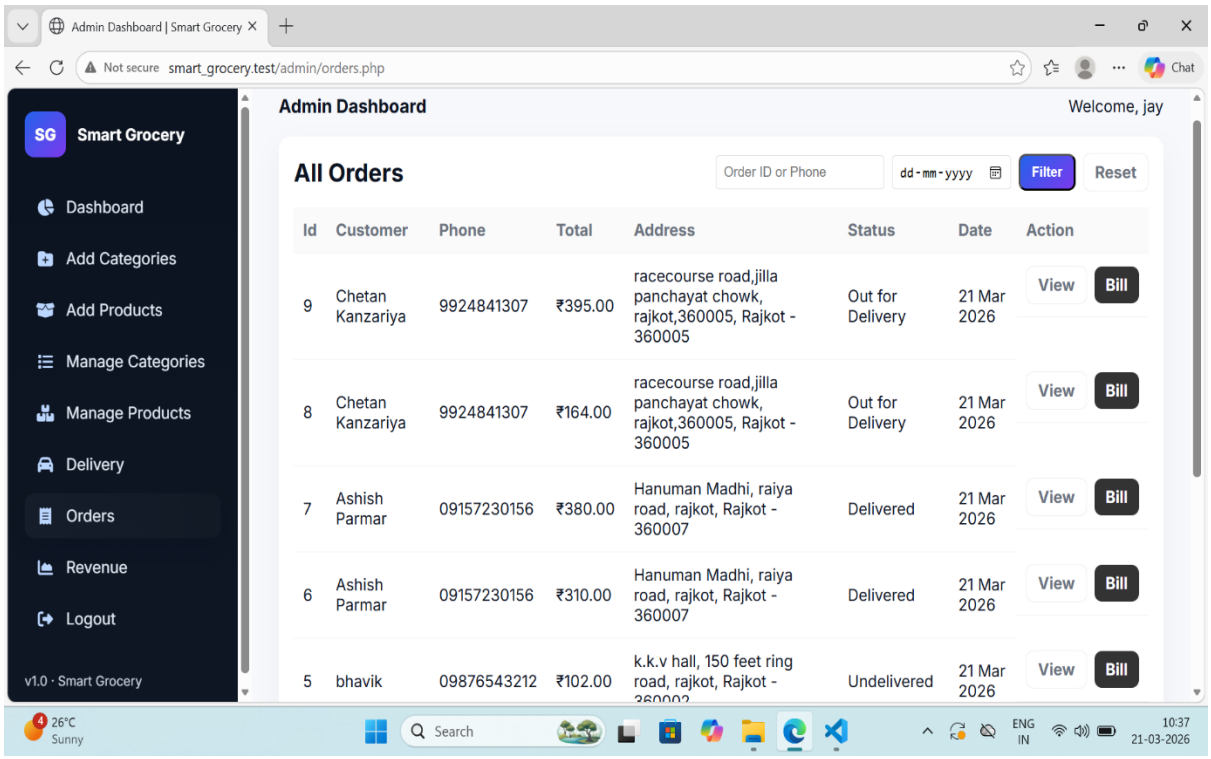
- Admin can Update and delete Products from manage Products page.

Assign Orders



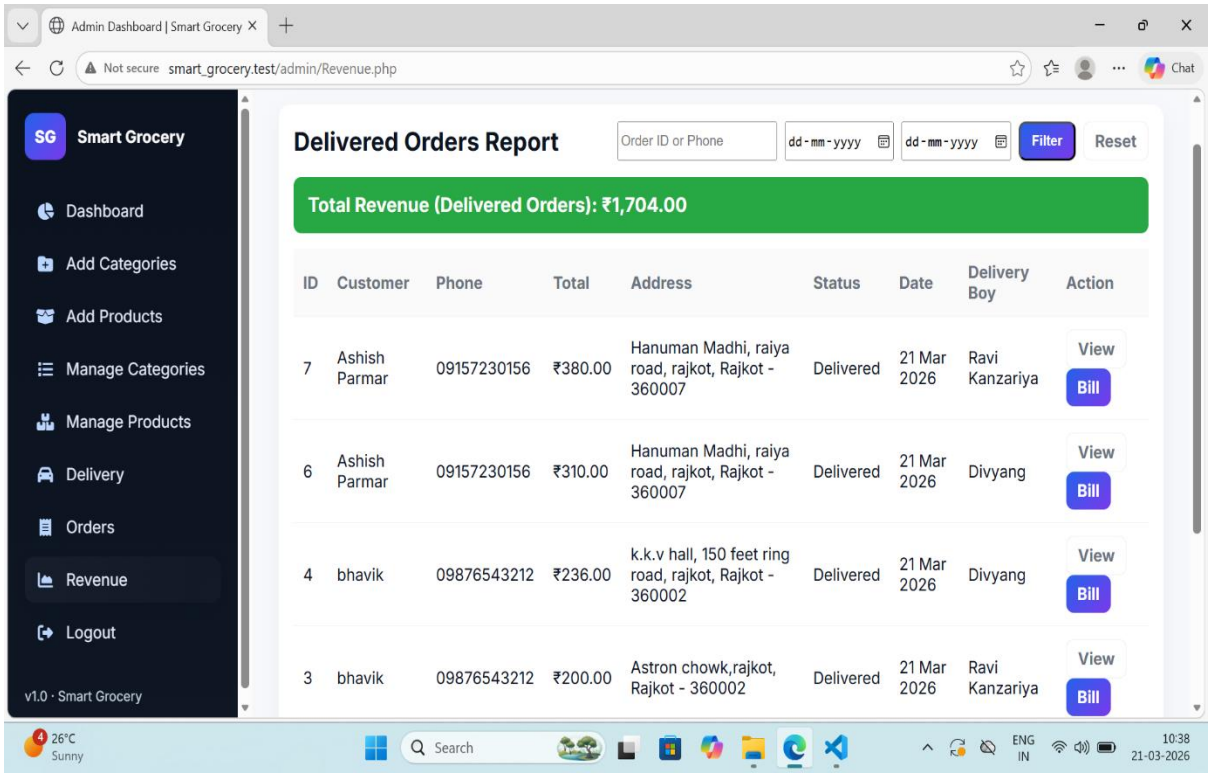
- Admin Can Assign Orders To Delivery Boy from Delivery page.

Orders



- Admin Can view all orders.
- Admin can Saw Order status from this page.
- Admin can generate Order bill.

Revenue Report



- Admin can Generate Revenue report of the selected time period from Revenue Report page.

Limitations

1. High Initial Cost

- Requires investment in:
  - Software development (PHP, database, admin panel)
  - Hardware (barcode scanners, POS, servers)
- Small businesses may find it expensive to implement.

2. Dependence on Internet & Technology

- System relies on: Internet connectivity, Server uptime
- If the system goes down → **orders, billing, and tracking stop completely**

3. Data Security Risks

- Stores sensitive data: Customer details, Payment information
- Vulnerable to: Hacking, Data leaks
- Needs strong security (encryption, authentication)

4. No AI or Recommendation System

- The system doesn't suggest products based on customer preferences or order history.

5. Logistics & Operational Complexity

- Managing: Orders, Deliveries, Returns
- Becomes difficult without automation

6. Delivery Management Challenges

- Issues in: Tracking delivery boys, Managing delivery time slots, Handling failed deliveries

7. Lack of Inventory management

- The system does not track ingredient stock or notify when an item runs out.

8. No Multiple Store Support:

- Designed for a single Store. Cannot handle multiple branches or locations.

9. Scalability Issues (Basic Systems)

- Simple PHP systems may:
  - Slow down with more users/orders
  - Need optimization for large-scale use

Conclusion

- The Smart Grocery Store System successfully automates the essential operations of a grocery business, including customer registration, product browsing, cart management, order placement, and administrative control. It provides a convenient and efficient platform for customers to purchase groceries online while enabling administrators to manage products, categories, orders, and customer data effectively.

- By implementing this system, the traditional manual shopping and billing process is simplified, reducing human errors and saving time for both customers and store owners. The system enhances the overall user experience through an intuitive interface and ensures better inventory and order management for the business.

- Although the current system meets the core requirements of an online grocery store, there is scope for further enhancements such as integration of online payment gateways, real-time order tracking, delivery management optimization, and advanced reporting and analytics features. Overall, this project demonstrates the practical implementation of PHP, MySQL, HTML, CSS, and JavaScript in developing a scalable, reliable, and user-friendly Smart Grocery Store application.

Feature Enhancement

1. Online Payment Gateway Integration

Integrate secure payment options such as Razorpay, PayPal, or Stripe to allow customers to make payments using credit/debit cards, UPI, and digital wallets for a seamless checkout experience.

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2. Real-Time Order Tracking

Enable customers to track their grocery orders in real-time with status updates like *"Order Confirmed"*, *"Packed"*, *"Out for Delivery"*, and *"Delivered"*.

---

3. Multi-Admin and Role-Based Access

Introduce multiple admin roles such as Store Manager, Inventory Manager, and Delivery Staff, each with specific permissions to improve system control and security.

---

4. Inventory and Stock Management

Implement real-time stock tracking to automatically update product availability, prevent overselling, and generate alerts when stock levels are low.

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5. Customer Feedback and Rating System

Allow customers to rate products and provide feedback, helping improve product quality and customer satisfaction.

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6. Delivery Slot Booking System

Provide customers with the option to select preferred delivery time slots for better convenience and efficient delivery planning.

---

7. Discounts and Coupon Management

Add features for promotional offers, discount codes, and special deals to attract new customers and retain existing ones.

8. Email & SMS Notifications

Send automated notifications for order confirmation, shipping updates, delivery status, and promotional offers via email and SMS.

9. Mobile Application Integration

Develop a mobile-friendly application for Android and iOS platforms to provide a smooth and accessible shopping experience.

10. AI-Based Recommendation System

Suggest products to customers based on their purchase history, preferences, and trending items to enhance sales and user engagement.

11. Multi-Store Support

Expand the system to handle multiple grocery store branches, allowing centralized management of products, orders, and inventory.

12. Customer Loyalty Program

Introduce reward points, cashback, or membership benefits to encourage repeat purchases and build customer loyalty.

13. Enhanced Security Features

Implement strong security measures such as password hashing, CAPTCHA verification, secure authentication, and data encryption to protect user information.

Webliography

- https://getbootstrap.com/
- https://stackoverflow.com/
- https://www.mysql.com/